

How a World Model Settles on a Future: Physics, Predictive Shortcuts, and Causal Control in Video Generation

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Abstract

Generative video world models learn from observations in which appearance, context, and dynamics are correlated, creating predictive shortcuts that fail under novel cue combinations. We study which training-compatible future the model generates. In a controlled appearance–dynamics setting, we hold physical history fixed, sweep only the appearance cue, and measure motion and appearance in fully decoded rollouts. Cue strength and physical history jointly determine the generated motion and appearance. Even when the shortcut wins, matched interventions recover the history-consistent future through a compact, state-conditioned family of edits. A fit-only controller synthesizes effective held-out edits from target direction and low-order boundary state without a held-out donor activation. We call a future causally writable while such edits can still redirect it, and committed once this access closes. Early writeability predicts which shortcut failures training later corrects, while the failures that remain become less writable. Relevant information persists after direct editing closes; condition-to-target self-attention K/V writes determine whether it controls decoded generation. In a pretrained 1.3B video DiT, the donor-free controller approaches the oracle edit, and learning cue-independent dynamics before biased post-training shifts a sharp commitment boundary later in the network. Physical failure can therefore reflect which future gains control even when a history-consistent prediction remains available.

1 Introduction

Generative world models learn to predict how observed scenes evolve (Ha & Schmidhuber, 2018; Hafner et al., 2020; Ho et al., 2022). Yet predictive factors in observational video rarely vary independently: appearance, context, identity, and dynamics tend to co-occur in regular ways. Highway scenes, for example, more often contain fast vehicle motion, whereas residential scenes more often contain slow motion. Context does not determine speed, but it is predictive in the data. A model may therefore predict displacement from recent motion or use scene appearance as a proxy. Both strategies can fit ordinary training examples, while uncommon appearance–dynamics combinations distinguish between them. When these cues conflict, which future does the model generate?

A failure under such a rare combination is internally ambiguous. The model may lack an internally recoverable history-consistent future, or that future may remain available while the shortcut controls generation. This is predictive underspecification: multiple solutions compatible with the training data can imply different futures under cue conflict (Geirhos et al., 2020; D’Amour et al., 2022).

We study this ambiguity in a controlled red–slow/blue–fast setting using matched cue sweeps that vary appearance while holding physical history fixed (Figure 1).

Training data can support a history-based predictor and an appearance shortcut that diverge under conflict. We ask whether the history-consistent future is absent or unused. Matched edits show that it can remain recoverable. Causal writeability measures how deep such edits remain effective; closure marks commitment. Training changes which errors are corrected and where commitment occurs. We then identify the downstream computation that lets the selected future control generation.

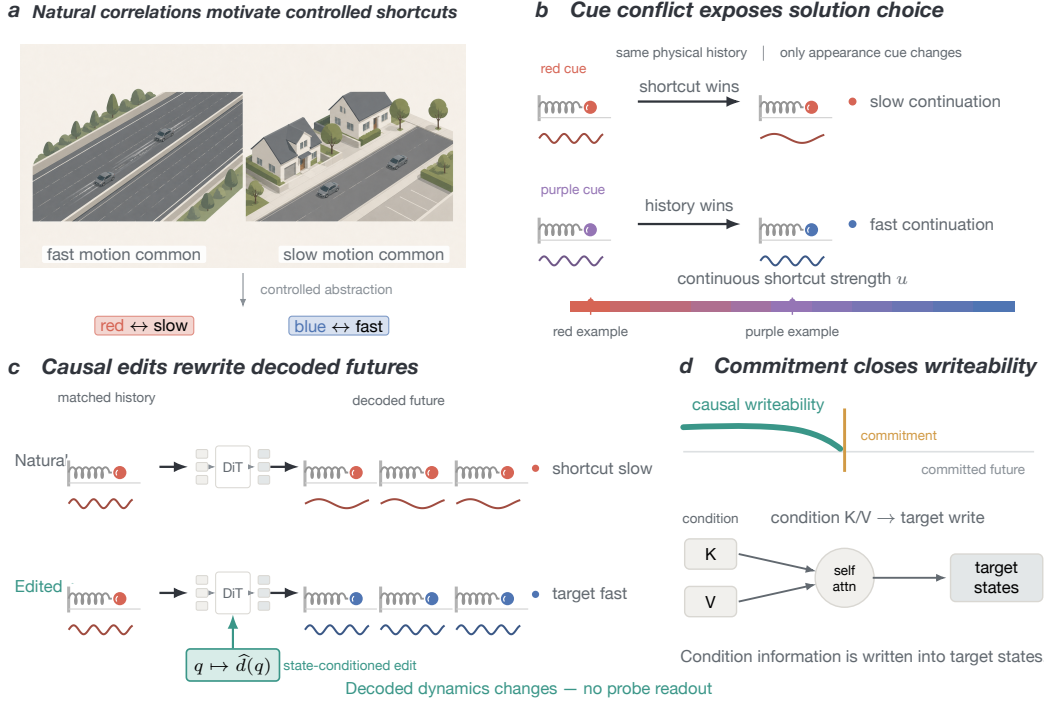


Figure 1: **The model may follow a shortcut even while the history-consistent future remains recoverable.** **a**, Natural context–motion correlations motivate a controlled color–dynamics shortcut. **b**, With history fixed, cue strength changes the generated future. **c**, A state-conditioned internal edit tests whether the history-consistent future remains recoverable. **d**, Writeability closes when condition-only edits can no longer redirect the future.

Contributions. We make three main contributions: (i) a decoded-rollout evaluation of how appearance and physical history compete to determine the generated future; (ii) a compact, state-conditioned family of edits together with causal writeability, an intervention-defined measure of how deep into the network a future can still be changed; (iii) evidence that the relevant information persists after direct editing closes, together with a downstream account of how commitment is implemented.

2 Which Future Does a World Model Choose?

We first ask how appearance and motion history compete to determine the generated future.

Experimental setting. We study Spring, a 488M-parameter latent flow-matching transformer with 30 bidirectional DiT blocks, 9 attention heads per block, and a frozen Wan2.1 VAE. Each clip shows a colored mass oscillating on a spring. Training pairs red with the slow band $\omega \sim U[2.2, 3.0]$ rad/s and blue with the fast band $\omega \sim U[5.2, 6.4]$ rad/s, creating an appearance–dynamics shortcut. Each 20-fps clip contains 65 conditioning frames and 64 generated frames at 128×128 ; the VAE compresses these to 17 condition and 16 target latent frames. No conflicting color–frequency combinations appear during training. We train two settings that differ in the amount of motion history available: Long retains all 65 conditioning frames, whereas Short replaces frames 0–56 with a fixed background before VAE encoding, leaving only the final eight motion frames while preserving the timeline and token count. Natural generation uses 20 flow-matching calls. We estimate oscillation frequency and appearance from the detected mass pixels in decoded futures. Edit, controller, and writeability experiments intervene on condition-prefix activations after a selected one of the 30 Transformer blocks at every flow-matching call, while leaving target-token activations unchanged at the intervention point. Appendix A gives the full contract.

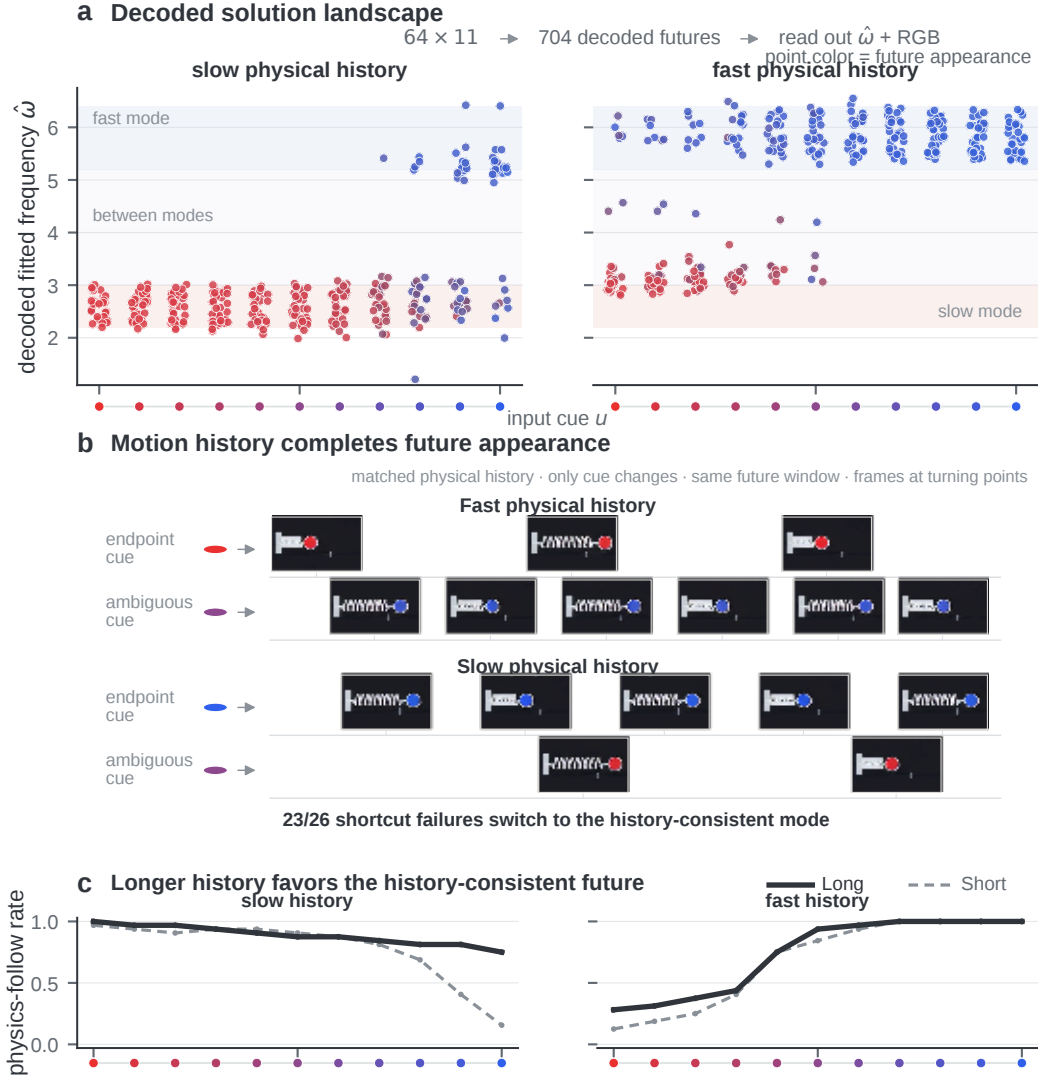


Figure 2: **Appearance and motion history compete to determine the generated future.** **a**, A matched hue sweep reads frequency and RGB from decoded futures. **b**, Under an ambiguous cue, motion history shifts both future motion and appearance toward its paired mode. **c**, Longer history resists stronger conflicting cues.

At test time, for each of 64 physical histories, we fix trajectory and generation seed while sweeping 11 hues from red to blue. The resulting 704 decoded futures (64 histories × 11 hues) reveal how the appearance cue changes the realized frequency $\hat{\omega}$ and appearance at fixed history (Figure 2a). Unlike a probe, this evaluation measures the dynamics the model actually generates.

Cue strength shifts the generated future. For a fixed slow history, increasingly blue cues make fast motion more likely. For a fixed fast history, increasingly red cues similarly favor slow motion. Around ambiguous hues, futures remain much more history-consistent than at the red/blue endpoints. Across the hue sweep, outputs concentrate around the two training-supported frequency bands, with trajectory-specific transitions and occasional compromise outcomes (futures at intermediate frequencies) near the crossover.

Motion history also determines future appearance. Under an ambiguous appearance cue, a fast history shifts motion toward the fast band and simultaneously pulls future appearance toward blue; a slow history produces the converse change (Figure 2b). Evidence in either variable therefore com-

pletes the associated motion–appearance pair as a unit. In a representative checkpoint, 23 of 26 histories that follow the shortcut at the red/blue endpoint return to the history-consistent frequency band under the ambiguous cue, while future appearance shifts toward the color paired with that motion during training.

More physical evidence shifts the competition. In separately trained Short and Long settings, history-consistent futures persist farther toward the red/blue endpoints under Long than under Short in both directions (Figure 2c). Thus the generated future depends jointly on appearance cue strength and available physical evidence. A controlled Pendulum replication shows the same evidence-dependent competition between appearance cue and physical history (Appendix B.6). Yet behavior alone cannot tell us why the shortcut wins. Is the history-consistent future absent, or can it still be recovered inside the model?

3 A Physical-State-Conditioned Edit Can Recover the History-Consistent Future

We next test whether the history-consistent future remains internally recoverable when natural generation follows the shortcut.

We report results from three independently trained Short 50K checkpoints. We form strict aligned–conflict pairs: trajectories sharing the same physical history, boundary state, and generation seed, where the aligned rollout has a history-consistent color (e.g., fast history with blue) and the conflict rollout has a shortcut color (e.g., fast history with red). The conflict rollout follows the shortcut, whereas its aligned counterpart follows history. Let $h_{C,i}$ and $h_{A,i}$ denote their condition states after DiT block ℓ . Adding the difference $d_i = h_{A,i} - h_{C,i}$ to the conflict run redirects its decoded future (Figure 3a).

Let $\hat{\omega}_{C,i}$ denote the fitted frequency of the natural conflict rollout, $\hat{\omega}_{A,i}$ that of its matched aligned rollout, and $\hat{\omega}_{\text{edit},i}(\ell)$ the fitted frequency after intervention at block ℓ . We measure normalized frequency recovery by

$$R_i^\omega(\ell) = \frac{\hat{\omega}_{\text{edit},i}(\ell) - \hat{\omega}_{C,i}}{\hat{\omega}_{A,i} - \hat{\omega}_{C,i}},$$

so $R^\omega = 0$ denotes the natural conflict frequency and $R^\omega = 1$ the matched aligned frequency. A layer scan selects a late site where the edit still redirects the decoded future; across the three runs this site falls at blocks 3, 4, and 6, with the intervention touching condition tokens only. The matched edit recovers the aligned frequency, but whether d_i belongs to a reusable edit family or is merely a donor-specific perturbation remains open.

The edit family is compact and state-conditioned. We fit an uncentered PCA basis on one set of matched differences and project held-out edits into it. Across the three Short 50K runs, retaining only four PCA coordinates preserves most of the executable effect: 87.8% of held-out top-4 edits satisfy $.75 < R^\omega < 1.25$, compared with 92.4% for the full matched-difference edit. Each run supports two rewrite directions (slow→fast and fast→slow), giving six run–direction groups; across these six groups, the median R^ω of the top-4 edit ranges from .910 to 1.012. Low rank rules out an arbitrary high-dimensional donor difference, but compactness alone does not explain what varies across trajectories. We therefore ask whether physical boundary state organizes the four coordinates. We use *route* below only as shorthand for this compact edit family.

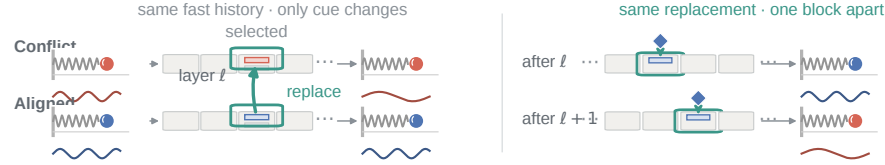
Boundary state provides a physical coordinate system for the edit. For harmonic motion, $x = A \cos \theta$ and $v = -A\omega \sin \theta$, so the input boundary phase

$$\theta^* = \text{atan2}(-v^*/\omega_{\text{true}}, x^*).$$

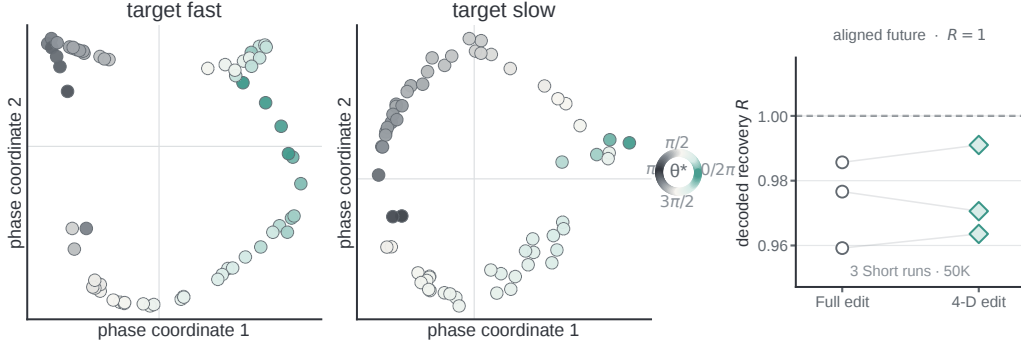
compactly encodes normalized position and velocity: up to a common amplitude scale, $\cos \theta^*$ is position-like and $\sin \theta^*$ is velocity-like after frequency normalization. Phase is the natural oscillator-specific coordinate; the general object is the physical state at the prediction boundary. The requested target direction selects the slow or fast frequency family, while the boundary position and velocity locate the trajectory-dependent edit within that family. In the four-dimensional edit coordinates, held-out trajectories form smooth phase-ordered loops in both rewrite directions (Figure 3b). We fit fast- and slow-target edits separately with a first-harmonic map,

$$\hat{z}_{1:4}^{(r)}(\theta^*) = \beta_0^{(r)} + \beta_c^{(r)} \cos \theta^* + \beta_s^{(r)} \sin \theta^*, \quad r \in \{\text{fast}, \text{slow}\},$$

a A matched state replacement recovers the history-consistent future



b Boundary position and velocity organize the compact edit



c A fit-only boundary-state controller synthesizes the edit without a donor

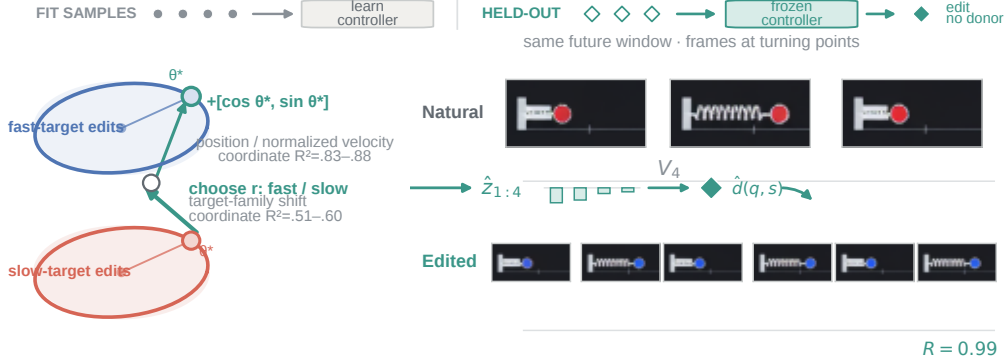


Figure 3: **A compact edit organized by interpretable physical state can recover the history-consistent future without a held-out activation donor.** **a**, A matched condition-state edit redirects a shortcut-selected future. **b**, Four coordinates retain nearly the full decoded effect, and held-out edits vary smoothly with boundary position and velocity. **c**, A controller predicts effective edits from target direction and boundary phase, then rewrites held-out futures without donor activations.

This physical-state model raises held-out coordinate R^2 to .83–.88, up from .51–.60 for a direction-only model. For a held-out receiver, the frozen controller maps the requested target and pre-existing boundary position and velocity through the frozen basis, without reading a held-out donor activation or generated future. Across the six run–direction groups, 85.9% of these synthesized held-out edits satisfy $.75 < R^w < 1.25$, with median R^w ranging from .910 to 1.009 (Figure 3c).

A complementary after-conflict controller obtains compatible edits from the decoded conflict state and a specified target using frequency plus position- and velocity-like differences (Appendix B.5). Together, the two parameterizations identify a low-order physical-state organization of the executable edit geometry. The edit jointly shifts motion and its training-associated appearance—for example, slow→fast also shifts future appearance toward blue—but succeeds only at some computational sites.

In Pendulum, angle and angular velocity organize top-four causal edit coordinates, and a fit-only prediction recovers held-out decoded dynamics (Appendix B.6).

4 Causal Writeability Reveals When Futures Become Committed

We then ask how long this internal access remains effective as computation proceeds through the network.

We apply the matched intervention at every site across the network. Early edits redirect the shortcut future; recovery then drops sharply over a narrow depth range (Figure 4a). Thus the decisive loss of direct condition-side control is localized to one or a few adjacent blocks rather than spread uniformly across the network. Individual boundaries occur at different depths, so averaging trajectories or runs can make this step-like closure appear broader. A future is causally writable while the edit can redirect it and committed once this direct access closes.

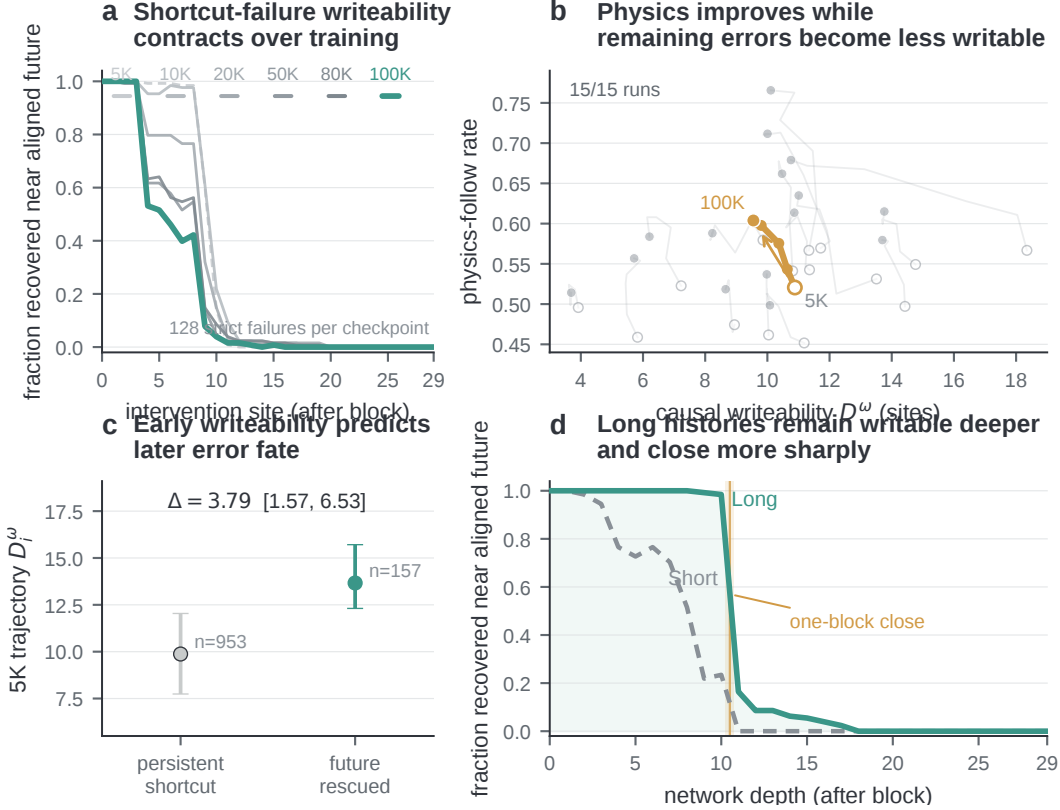


Figure 4: **Direct editability closes with depth and predicts which early failures training later corrects.** **a**, Strict-failure banks are rebuilt at each checkpoint; layerwise recovery drops earlier over training. **b**, Physics behavior improves while writeability contracts within every run. **c**, At 5K, future-rescued failures are more writable than persistent failures. **d**, Long histories preserve writeability deeper and delay closure.

For a strict-failure bank (pairs in which the aligned rollout follows the physical history and the conflict rollout follows the color shortcut), define

$$D^\omega = \sum_{\ell \in \mathcal{L}} \Pr_i[\text{valid} \wedge .75 < R_i^\omega(\ell) < 1.25],$$

over the common 31-site scan $\mathcal{L}$ (one site before block 0 and one after each of the 30 blocks). D^ω is the expected number of sites supporting a near-aligned rewrite. Here valid means that the edited video yields a usable decoded frequency fit; invalid edits do not count as writable. D^ω measures causal revisability, not representation depth.

Training separates error correction from error consolidation. This resembles a margin effect: some ambiguous examples cross into the correct region, while unresolved examples move farther

into the region supporting the same wrong prediction. Across all 15 Short runs, physics-following behavior rises from .52 at 5K to .60 at 100K, while mean writeability falls from 10.88 to 9.55 sites (Figure 4b). Training therefore improves average physical behavior while making the surviving failures less corrigible. A complementary comparison across runs at matched checkpoints is reported in Appendix B.8. This interpretation predicts that, among behaviorally identical early failures, the more writable ones should be the ones training later corrects.

Mechanistic divergence precedes behavioral divergence. We take trajectories that are strict short-cut failures at 5K and group them only by later fate: future-rescued trajectories are physics-following at both 80K and 100K, whereas persistent failures follow the shortcut at all six checkpoints (5K, 10K, 20K, 50K, 80K, 100K). For trajectory i ,

$$D_i^\omega = \sum_{\ell \in \mathcal{L}} \mathbf{1}[\text{valid} \wedge .75 < R_i^\omega(\ell) < 1.25].$$

Future-rescued errors are already writable at 3.79 more sites on average (95% CI [1.57, 6.53]; Figure 4c). Early writeability thus distinguishes the later outcomes of behaviorally identical failures (Appendix B.7).

More physical evidence delays commitment. In all three matched-seed 50K comparisons, the Long setting remains writable to a later block than the Short setting (Figure 4d); the displayed Long run closes within a single block. Commitment depth therefore depends on the amount of physical evidence, not architecture alone.

Training path selects which future gains control. To test whether this dependence extends beyond the from-scratch Spring model, we adapt a pretrained Wan 1.3B video DiT. The Direct path learns the biased red-slow/blue-fast task immediately. The Neutral path first learns the same Spring dynamics from achromatic videos, where appearance is uninformative, and then receives exactly the same biased post-training. On a newly frozen, unbiased population of 128 physical histories, conflict physics-follow at 10K is 80.5% after neutral adaptation and 9.4% under direct adaptation, while aligned performance is matched (93.8% versus 96.1%; Figure 5a). The donor-free phase controller reaches median recovery $R^\omega = .953/.923$ for fast/slow edits, close to the oracle top-four reference .968/.952 (Figure 5b). On the same persistent-failure identities, Direct adaptation closes between B12 and B13, whereas Neutral-first closes between B14 and B15 (Figure 5c). Thus training history moves a sharp commitment boundary while preserving the same compact edit structure. In an exploratory fixed-identity audit, intermediate-layer access can also close and reopen over nearby training checkpoints. Commitment is therefore checkpoint-local and can reopen during optimization (Appendix B.3).

These analyses identify when direct condition-side access closes. Does commitment erase the relevant information, or change where it controls generation?

5 Condition-to-Target Writes Realize Commitment

Finally, we test what changes when direct condition-side editing stops working.

Commitment does not erase the relevant information. To measure the aligned target-state response, we first apply an edit at an earlier writable block. At each later block, we measure how the future-token activations change relative to the unedited conflict run, and project this change onto the difference between the matched aligned and conflict runs at the same block. A response of 1 means that the full aligned-conflict difference is reproduced along this direction, whereas 0 means that none of it remains. On held-out fast-target receivers in each of three runs, an aligned target-state response remains measurable from the first non-writable block through the final block, even after condition-side edits cease to change frequency (Figure 6a).

The edit coordinates transfer across independently trained models. At each model’s own rewrite site, fit pairs determine a global scale and a four-dimensional orthogonal map from source to target coordinates. Applied to held-out receivers, all six directed maps transfer controller-predicted edits with $R = .94-.99$, without consulting donor activations (Figure 6b). Models with different natural behavior can therefore share compatible edit coordinates. Their natural behavior differs in how downstream computation uses this information.

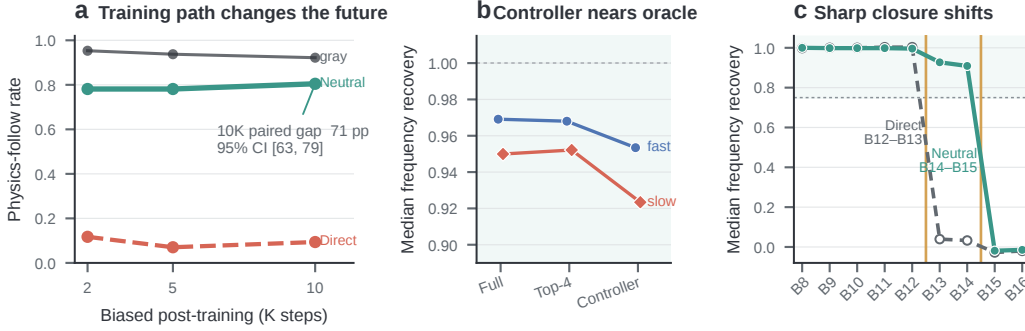


Figure 5: **Learning the dynamics before introducing the shortcut changes which future later controls generation.** **a**, Direct and neutral-initialized models are evaluated on the same 128 held-out histories; the large conflict gap persists from 2K to 10K, while aligned performance is matched. **b**, In the pretrained model, the donor-free controller approaches the full and oracle top-four edits. **c**, Direct and Neutral-first adaptation both close sharply, but at B12–B13 and B14–B15, respectively.

Self-attention writes the information into target states. Starting from a successful matched condition edit $E = C + (A - C)$, we restore one natural-conflict attention component or component set at a time. Restoring condition Q has little effect, whereas restoring condition K+V removes recovery in all six run-direction groups (Figure 6c). Patching the target stream gives the same localization: restoring the natural-conflict post-attention residual destroys recovery, whereas restoring the corresponding MLP output does not. These interventions place the necessary computation in self-attention, where target queries read condition K/V. Once that information has been written into target residuals, condition-only edits no longer redirect the rollout.

Different heads and components implement the same effect across runs. The write is concentrated in one K head in one model, distributed across K/V in another, and localized to a selective V bottleneck in the third (Appendix B.12). In that third model, we amplify only condition-token V in one of nine heads:

$$V_C^{(8)} + \lambda \Delta V^{(8)}, \quad \Delta V^{(8)} = V_M^{(8)} - V_C^{(8)},$$

where M denotes the controller-edited state produced by the frozen state-conditioned controller from § 3. At $\lambda = 8$, mean recovery reaches 1.05 and the physics-following rate peaks at 56.3% among 48 held-out strict failures (Figure 6d). Stronger writes overshoot: at $\lambda = 16$, all outputs remain valid, but 64.6% fall between or outside the supported bands. The relevant edit direction can therefore be present yet too weak at one downstream bottleneck to govern generation.

Timing also matters for this bottleneck: early flow-matching calls produce an internal response without changing the video, whereas sustained late writes recover the decoded future (Appendix Figure 25).

6 Discussion and Limitations

These results reframe how physical failures in generative world models should be interpreted. Under predictive underspecification, failure does not imply that a history-consistent future is absent. The relevant question is whether an effective internal edit remains available and whether downstream computation acts on the same information. Training can improve overall physics-following behavior even as the failures that remain become less revisable. Natural behavior therefore reflects which future is generated and when it becomes committed, beyond what a probe can read from activations.

Our mechanistic analysis relies on controlled synthetic dynamics, where exact counterfactual matching and quantitative decoded-video evaluation are possible. Because the edit changes motion together with its training-associated appearance, it should not be interpreted as a disentangled physics subspace. Causal writeability measures direct condition-side revisability; it should not be conflated

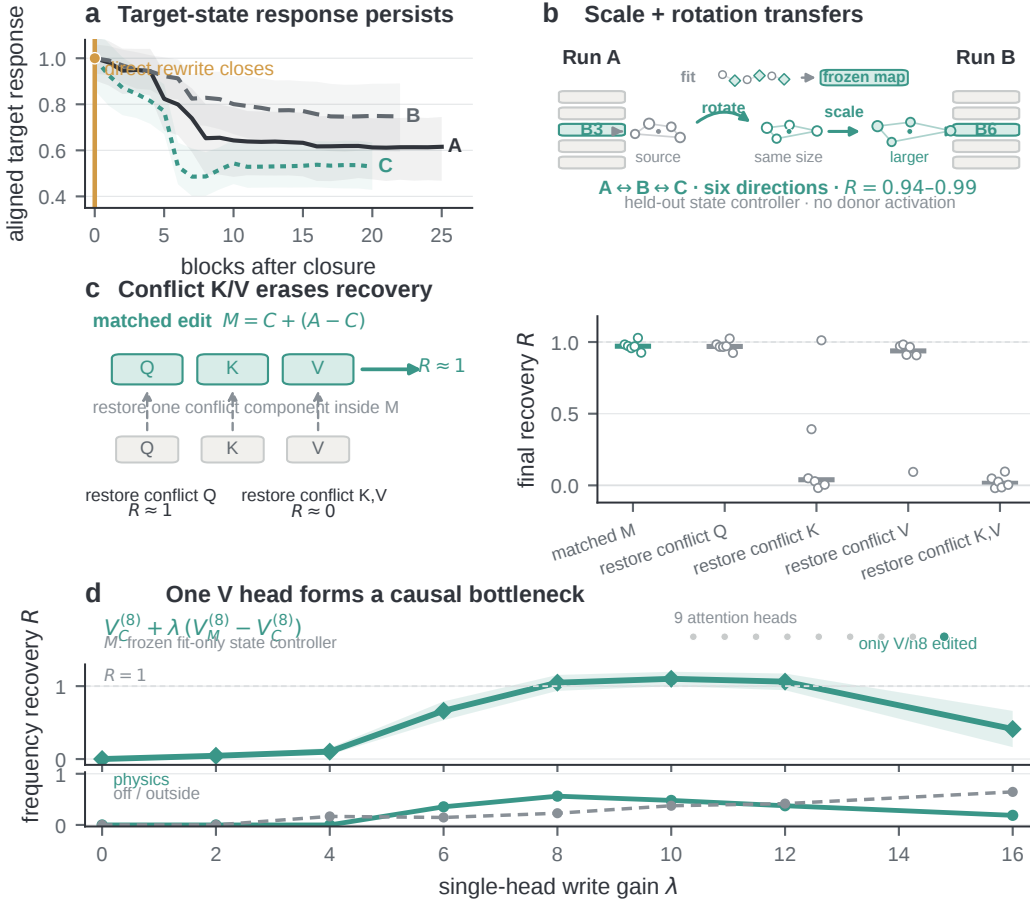


Figure 6: The relevant information persists after direct editing closes and gains control through attention writes into target states. **a**, In a focused fast-target audit, aligned target-state responses persist after condition-side rewriting closes. **b**, A scaled orthogonal alignment transfers causal coordinates between independently trained models at their own rewrite sites. **c**, Restoring conflict K+V, but not Q, removes recovery and localizes the target write to self-attention. **d**, Amplifying one identified V head reveals a finite margin; excessive gain leaves the supported modes.

with representation depth or irreversibility. The exact K/V components and attention heads responsible for the write also differ across runs. Our controlled Wan adaptation shows that training history can redirect a pretrained solution, but whether the same commitment dynamics govern naturally occurring shortcuts in broadly pretrained world models remains open.

7 Related Work

Shortcut learning and predictive underspecification. Training data can support multiple predictive rules that agree on the training distribution and diverge only under shift (Geirhos et al., 2020; D’Amour et al., 2022). Closest to our behavioral setting, Kang et al. (2025) study held-out combinations in controlled mechanics videos and find case-based generalization with an attribute hierarchy that places color above velocity. That work asks which observed attribute is prioritized under held-out combinations. We ask which future is generated under predictive underspecification. A matched continuous cue sweep, joint appearance–dynamics readout, and Short–Long evidence control show evidence-dependent selection governed jointly by shortcut strength and physical history. Behavioral priority alone cannot distinguish a history-consistent future that was never learned from one that remains internally available but unused.

Physics representations and control in video models. Recent work establishes that physics-relevant structure can be decoded, localized, and steered in video models. Joseph et al. (2026) identify distributed physical-variable geometry and a Physics Emergence Zone in video encoders; Esmati et al. (2026) decode physical plausibility along inverted video-diffusion trajectories; and Alam (2026) steer plausibility judgments with concept activation vectors. At the generator level, VideoREPA injects relational physics during training, while WMReward guides inference-time generation with a latent-world-model plausibility reward (Zhang et al., 2025; Yuan et al., 2026). These studies establish the presence or controllability of physics-relevant structure. We study its control over decoded generation: whether an internally recoverable future governs the fully decoded output, how long it remains writable, and how that control is implemented downstream.

Causal intervention, commitment, and cross-model transfer. Activation patching and causal tracing localize causal influence (Meng et al., 2022; Zhang & Nanda, 2024). Subspace-patching work distinguishes causal access from the computation used naturally (Makelov et al., 2024); we make the same distinction. Manifold steering links activation geometry to behavioral trajectories, including physical dynamics in a video world model (Wurgaft et al., 2026). Most directly for commitment, Plattner et al. (2026) localize a 3D-diffusion failure to a cross-attention write near an early reverse-process bifurcation. Model stitching and re-basin methods show that independently trained networks can be aligned (Bansal et al., 2021; Ainsworth et al., 2023), while functional alignment alone need not imply shared information (Smith et al., 2025). Learned mappings have also transferred activation interventions across models and modalities (Oozeer et al., 2025; Poppi et al., 2026). Our test is narrower and causal: a fit-only alignment must transfer state-conditioned edits and reproduce their effect in decoded video. Our contribution is therefore not a new patching primitive, but an account of how state-conditioned edits that transfer across models gain or lose control over a generated future.

Reproducibility statement

Appendix A documents the model, data generation, decoded evaluator, statistical units, strict-pair construction, and intervention protocol. We will release training and evaluation code, configurations, data generators, random seeds, checkpoint metadata and hashes, frozen evaluation manifests, frozen matched-pair identities and input frames, figure builders, and audit scripts. The manifests record the sample identities used in each comparison and link quantitative figures to their source artifacts. A release script reruns the evidence checks, verifies provenance, rebuilds the mechanism figures, and compiles the manuscript. The release will include all hyperparameters and experiment-specific settings needed to reproduce the reported results.

AI use statement

The authors used generative AI assistants, including OpenAI ChatGPT and Codex-based coding agents, to help refine hypotheses and experimental designs; implement and refactor code; run and monitor experiments; construct audits; analyze results; design figures; search the literature; and draft, translate, and edit the manuscript. The authors formulated the research questions, made all final methodological and interpretive decisions, approved each experiment, supervised its execution, inspected raw outputs, and reviewed the code, citations, figures, and claims in the submission. All reported measurements come from the stated deterministic evaluators and intervention pipelines, not language-model judgments. The authors reviewed all AI-assisted material and take responsibility for the final paper.

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A Experimental Details

Model and training. The Spring model is a no-text latent flow-matching transformer with 488,158,912 trainable parameters, excluding its frozen VAE. Its denoiser has 30 bidirectional DiT blocks, hidden width 1152, feed-forward width 4608, nine attention heads of width 128, and a $1 \times 2 \times 2$ latent patch size. A frozen Wan2.1 VAE maps each 129-frame video to a $33 \times 16 \times 16$ latent grid with 16 channels: 17 conditioning frames followed by 16 target frames. Spatial patching gives 1,088 condition and 1,024 target tokens. The two history regimes use the same architecture, token count, prediction boundary, and optimizer; they differ only in the visible physical history. Table 1 gives the remaining shared contract.

Table 1: **Spring experimental contract.** Long and Short use the same video span and latent layout; Short replaces early motion with the fixed background before VAE encoding.

Component	Setting
Training support	2,048 videos per regime: 1,024 red–slow and 1,024 blue–fast; no conflicting color–frequency combinations
Physical variables	slow $\omega \sim U[2.2, 3.0]$ rad/s; fast $\omega \sim U[5.2, 6.4]$ rad/s; normalized horizontal-displacement amplitude $\sim U[0.10, 0.17]$; phase $\sim U[0, 2\pi)$
Video and boundary	128×128 , 20 fps; frames 0–64 are conditioning frames and frames 65–128 are generated
Long / Short history	Long exposes real frames 0–64; Short replaces frames 0–56 by RGB (28, 30, 34) and exposes real motion in frames 57–64; replacement occurs before VAE encoding
Renderer	equilibrium $x = 0.58$, anchor $x = 0.16$, center $y = 0.50$, mass radius 7 px, 10 spring coils; red RGB (235, 48, 48) and blue RGB (48, 96, 235). Oscillator position x , amplitude, and boundary position x^* are expressed as normalized horizontal displacements from equilibrium; v^* is the corresponding displacement velocity per second. A displacement x is rendered at horizontal pixel coordinate $(0.58 + x)(W - 1)$
Latent sequence	33 latent frames and 2,112 tokens: 17 condition frames (1,088 tokens) and 16 target frames (1,024 tokens); only target frames enter the loss
Optimization	AdamW with PyTorch ConstantLR, batch size 32, gradient accumulation 1, base learning rate 2×10^{-4} , weight decay 0.01, no EMA
Training schedule	primary mechanism analyses use the 50K checkpoints; the 15-run longitudinal runs train to 100K and are evaluated at 5K, 10K, 20K, 50K, 80K, and 100K
Sampling	20 flow-matching calls, scheduler shift 5.0, denoising strength 1.0, and fixed per-trajectory generation seeds

Decoded-video evaluation. Evaluation uses generated pixels. A deterministic detector tracks the mass center in every future frame without reading the input hue label. Validity requires detection in at least 90% of frames, no missing run longer than three frames, adjacent and boundary jumps at most 12 px, vertical deviation at most 7 px, multi-component rate at most 0.10, and valid area and image bounds. For valid tracks, detected horizontal pixel coordinates are converted to normalized

displacement from equilibrium. We fit the detected future-frame positions at their original time points, without interpolating missing detections, using

$$x(t) = a \cos(\hat{\omega}t) + b \sin(\hat{\omega}t) + c$$

over a 1,601-point grid with $\hat{\omega} \in [1.21, 8.64]$ rad/s. Fits with RMSE at most 0.035 are retained; the RMSE is measured in the normalized displacement coordinate. Frequency-band membership uses a 0.005 rad/s tolerance at the slow and fast band boundaries. We measure future appearance from the mean RGB of the detected mass pixels across valid future frames—not by averaging entire video frames—and project it onto the red-to-blue color axis. The 11 cues linearly interpolate between RGB (235, 48, 48) and (48, 96, 235). A valid rollout is labeled physics (P) when $\hat{\omega}$ lies in the history-consistent band, shortcut (S) when it lies in the cue-associated opposite band, compromise (C) when it lies in the unsupported gap, and off-family/invalid (O) otherwise. The frozen 64-history $\times$ 11-hue response grid keeps trajectory identity, physical parameters, renderer, and generation seed fixed across hue. Consequently, hue variants are repeated measurements of 64 histories, not 704 independent samples.

Strict matched pairs. A strict pair consists of an aligned rollout that is valid and in its true frequency band and a conflict rollout that is valid and follows the opposite shortcut band. The pair shares trajectory, frequency, amplitude, phase, boundary state (x^*, v^*) , renderer, and generation seed; only mass color changes. The route and cross-run analyses use a shared 256-trajectory bank for the three 50K Short runs: 128 fast-target and 128 slow-target pairs, split into 128 fit and 128 held-out pairs balanced by direction. Every pair satisfies the strict criteria in all three runs and has $|\hat{\omega}_A - \hat{\omega}_C| \geq 2.0$. Writeability uses each seed-checkpoint’s local bank of natural strict failures, because that estimand asks which errors remain at that point in training. The persistent-error control freezes trajectory identities across checkpoints to test bank-composition effects directly.

Intervention operators. Layer localization and writeability replace the condition state,

$$h_{\ell, \text{cond}}^C \leftarrow h_{\ell, \text{cond}}^A,$$

whereas route analyses add a full, projected, transferred, or predicted edit,

$$h_{\ell, \text{cond}}^C \leftarrow h_{\ell, \text{cond}}^C + \tilde{d}_\ell, \quad d_\ell = h_{\ell, \text{cond}}^A - h_{\ell, \text{cond}}^C.$$

Both operators modify only the 1,088-token condition prefix at one site, hit that site once on each of 20 flow-matching calls, and leave the 1,024-token target suffix exactly unchanged at injection; the suffix then evolves normally through subsequent computation. FM-window and component-restoration experiments state their deviations explicitly. Layer scans cover 31 sites: the pre-block residual and after-B0 through after-B29. Route localization chooses the latest after-block at which at least 75% of edits in each direction are valid and satisfy $0.75 < R_i^\omega < 1.25$; this functional site is distinct from D^ω , which summarizes writeability over the full scan.

Statistical units and provenance. Figure 2 treats the 64 physical histories (32 per frequency band) as units and the 11 hues as repeated measurements. Route, controller, and transfer fits use trajectory-disjoint held-out sets. The cross-solution analysis resamples complete six-checkpoint seed trajectories; the future-fate analysis first resamples seeds and then trajectories within seed $\times$ direction $\times$ fate strata. Exact checkpoint, bank, split, bootstrap seed, validity record, and source-artifact hash are frozen, where applicable, in figure-specific machine-readable manifests.

B Supplementary Results

Throughout this section, Runs A, B, and C denote the independently trained Short models with seeds 3407, 3408, and 3409, respectively.

B.1 Behavioral robustness and matched cue reversal

Table 2 conditions on trajectories for which the strongly conflicting endpoint cue already produces the shortcut-associated frequency: red for a true-fast history and blue for a true-slow history. It then changes only the cue to purple ($u = 0.5$) for the same physical trajectory and generation seed. Thus

Table 2: **Endpoint shortcut failures often return toward the history-consistent continuation at the ambiguous cue.** Purple outcomes are counts within each endpoint shortcut cohort. P: physics; S: shortcut; C: compromise; O: off-family/invalid.

Run / history	Endpoint S cohort	Purple P	Purple S	Purple C	Purple O
A / fast	10/32	8	0	1	1
A / slow	16/32	15	0	1	0
B / fast	16/32	9	1	5	1
B / slow	8/32	8	0	0	0
C / fast	8/32	1	0	5	2
C / slow	9/32	9	0	0	0

the table measures a within-trajectory change in the naturally selected decoded continuation, not accuracy on two unrelated sample sets.

The categorical reversal is solution- and direction-dependent, especially for the fast-history branch of Run C. Nevertheless, over the endpoint-shortcut cohort, every run-direction group shifts both frequency and future color toward the history-associated joint mode on average: the mean frequency shift ranges from 1.68 to 2.76 rad/s and the normalized color shift from .50 to .71.

Figure 7 expands the decoded-video evaluation to all 15 Short runs at 50K. Each run uses the same 64 physical histories, 11 hues, and deterministic evaluator; all 10,560 rollouts are valid. The aggregate allocation varies substantially across runs, particularly in how often an unsupported compromise is realized. The cue-response panels nevertheless show the same evidence-dependent pattern: slow histories are most often preserved near red cues, fast histories near blue cues, with solution-specific transition widths around the ambiguous region.

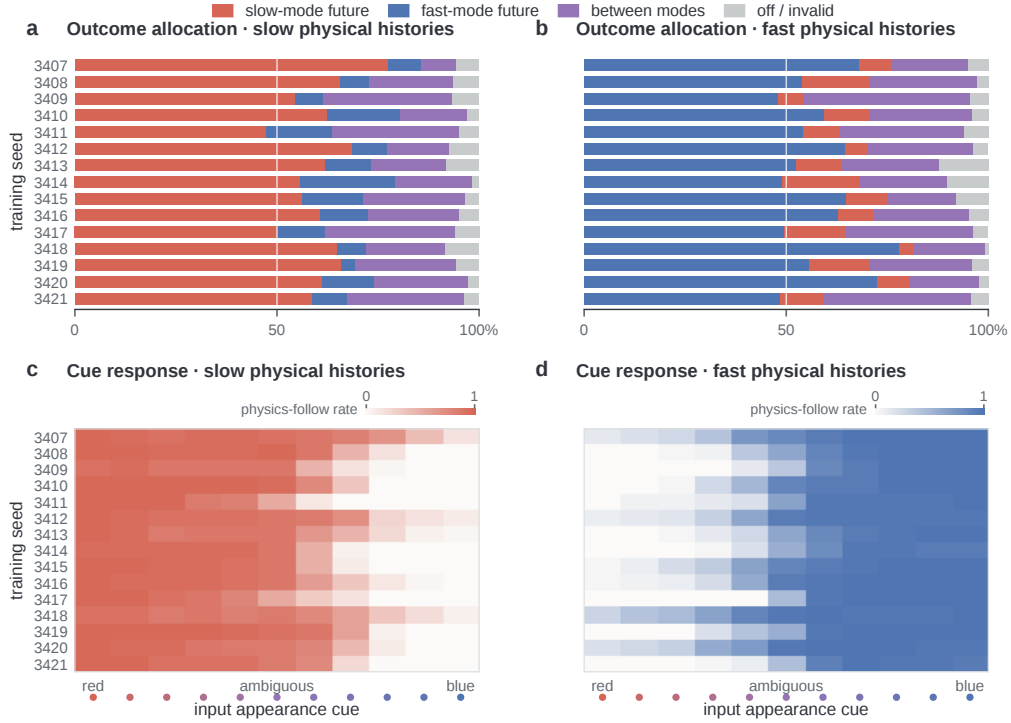


Figure 7: **Evidence-dependent joint-mode selection recurs across 15 independently trained models.** **a,b**, Outcome allocation over all 352 hue-conditioned rollouts in each physical band. Colors denote decoded motion modes, not correctness: coral is slow, blue is fast, purple lies between the supported bands, and gray is off-family or invalid. **c,d**, Physics-follow rate for each seed and cue; every cell averages 32 matched physical histories.

B.2 Pretrained Wan replication of the executable route and sharp closure

We repeat the route analysis in a pretrained Wan 1.3B video DiT adapted to the biased Spring task. A direction-specific first-harmonic controller is fit on 128 trajectories and frozen before evaluation on 128 disjoint receivers. The held-out coordinate fits reach $R^2 = .858/.771$ for fast/slow targets. In decoded rollouts, the donor-free controller attains median recovery $R^\omega = .953/.923$, close to the oracle top-four ceiling of $.968/.952$; neither controller fitting nor held-out intervention reads a matched donor activation.

The same model family also exhibits localized closure. On the same 17 slow-target persistent failures, direct adaptation loses median recovery between B12 and B13, whereas neutral-first adaptation retains it until B14 and loses it between B14 and B15. Thus the training path shifts the boundary without turning the loss of direct condition-side control into a gradual network-wide decay.

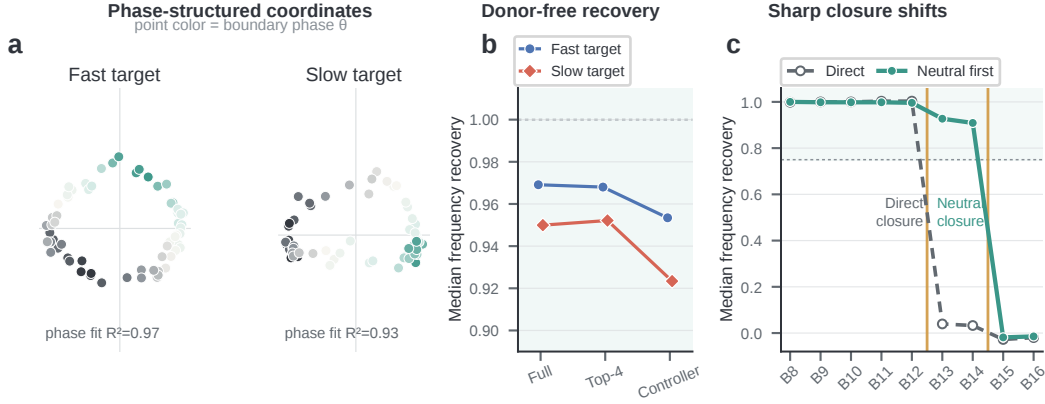


Figure 8: **A pretrained video DiT reproduces the executable route and localized commitment boundary.** **a**, Held-out top-four coordinates vary smoothly with boundary phase in both rewrite directions ($n = 64$ each; the displayed state-varying coordinate pairs are centered for visualization). **b**, The top-four oracle projection retains the full matched-edit recovery, and the fit-only phase controller approaches that ceiling without a held-out donor. **c**, Median recovery on the same persistent-failure identities drops across one adjacent-block transition under both training paths, while neutral-first adaptation shifts the transition later. The amber lines mark the two observed closure intervals; the shaded region denotes $.75 < R^w < 1.25$.

B.3 Pretrained neutral-to-biased curriculum

We initialize a Wan 1.3B video DiT in two ways. The Direct path adapts the pretrained checkpoint directly on the red–slow/blue–fast Spring data. The Neutral path first trains for 10K updates on 2,048 achromatic Spring videos, with slow and fast dynamics equally frequent and all masses rendered RGB (92, 92, 92), then starts the same biased stage from a fresh optimizer. The biased-stage data order and random streams are matched between paths. Gray and biased training manifests have disjoint trajectory identities.

Population results use a frozen evaluation set generated before model evaluation: 128 physical histories (64 per frequency band), each rendered at 11 red-to-blue hues and one gray appearance. Its base seeds are disjoint from both training manifests, the earlier development hue set, and the mechanism cohort. Table 3 reports endpoint behavior with invalid outputs included in O. Every biased-checkpoint endpoint and gray rollout is valid.

Table 3: **Neutral adaptation redirects the solution reached under the same subsequent biased data.** P/S/C/O are decoded conflict outcomes on the same 128 held-out histories. The paired interval resamples physical histories.

Biased step	Path	Aligned P	Conflict P	S	C	O	Neutral–Direct P
2K	Direct	83.6	11.7	61.7	16.4	10.2	
	Neutral	93.8	78.1	12.5	7.8	1.6	66.4 [57.8,74.2]
5K	Direct	93.0	7.0	65.6	23.4	3.9	
	Neutral	96.1	78.1	7.0	14.8	0.0	71.1 [63.3,78.9]
10K	Direct	96.1	9.4	58.6	28.1	3.9	
	Neutral	93.8	80.5	7.8	11.7	0.0	71.1 [63.3,78.9]

At the achromatic 10K starting checkpoint, the median matched red–blue endpoint frequency difference is .056 rad/s and opposite-band shortcut outcomes are absent at the nominal conflict endpoint. The checkpoint is an imperfect Spring generator, especially for fast histories, but hue does not yet control the generated frequency systematically. Within the Neutral biased path, matched gray inputs remain more physics-following than colored conflicts by 17.2, 15.6, and 11.7 percentage points at 2K, 5K, and 10K, respectively.

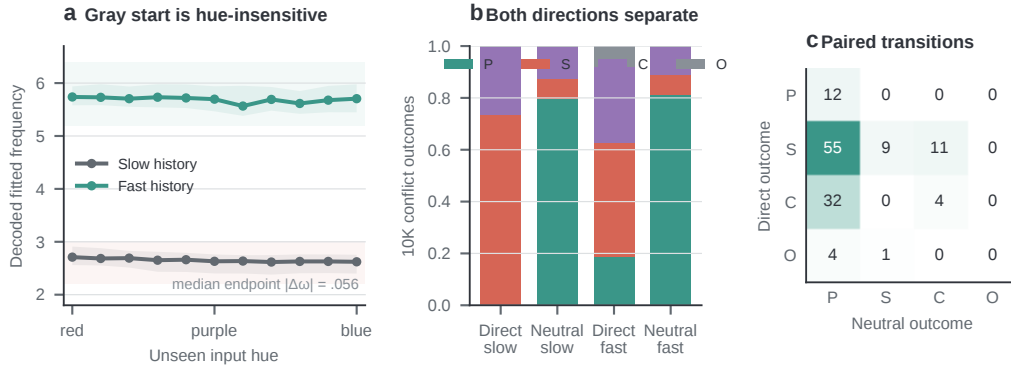


Figure 9: **A cue-independent starting solution redirects biased post-training on a frozen held-out population.** **a**, Before biased training, median decoded frequency changes little over unseen hue; shading is the interquartile range across histories. **b**, The 10K conflict difference occurs in both physical directions. **c**, Counts for the same 128 histories show that most Direct shortcut/compromise outcomes become physical under Neutral initialization. The post-training population curves already shown in Figure 5a are not repeated here.

The event-centered causal analysis uses a separate seed-disjoint bank enriched for shortcut-susceptible histories and therefore does not estimate population failure prevalence. At 2K, 17 slow-direction failures already present at 1K have median deepest writable block B12, whereas ten failures newly observed at 2K have median B15 (difference three blocks, trajectory-bootstrap 95% CI [2, 4]). Among all 27 slow strict failures at 2K, those still shortcut-selected at 3K have median boundary B12, versus B14.5 for non-persistent failures (difference 2.5 blocks, 95% CI [2, 3]).

The 2K depth ordering remains strong for behavior measured at 4K and 5K but attenuates by 10K, supporting a near-term stabilization signal whose predictive power weakens over longer horizons. On 11 fixed identities, recovery at B13/B14 also closes and reopens over nearby checkpoints while B12 remains writable and B15 remains typically closed. This exploratory fixed-identity result makes commitment a checkpoint-local computational property, not an absorbing state across further optimization.

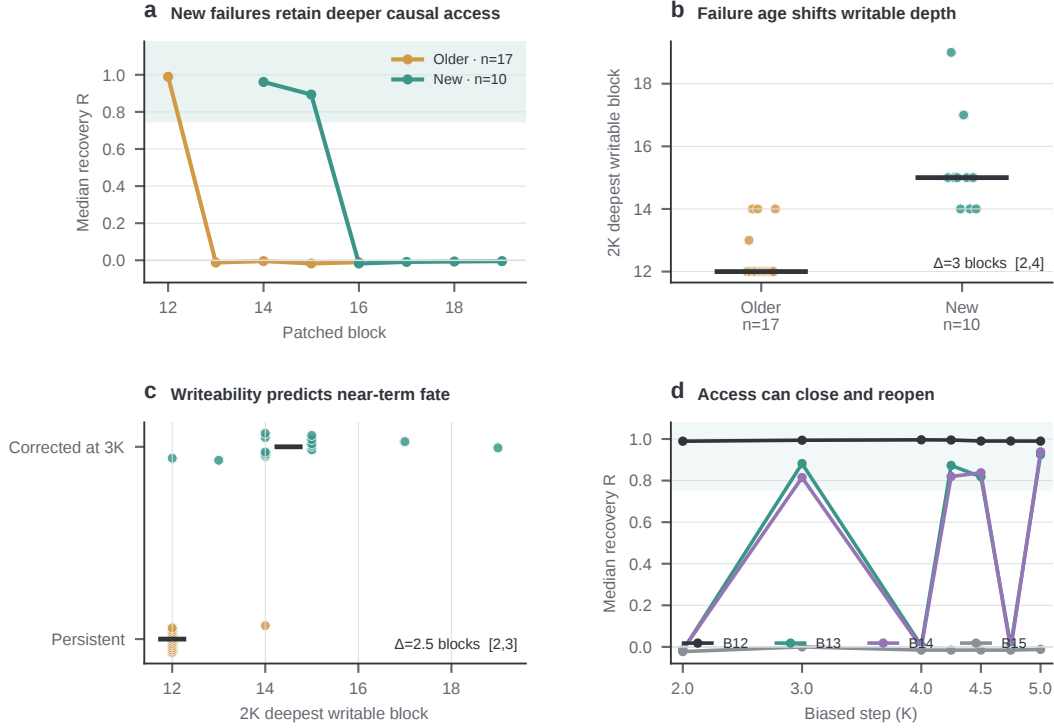


Figure 10: **Behavioral failure precedes stable causal closure in the Neutral path.** **a**, At 2K, newly observed failures retain effective edits about three blocks deeper than failures already present at 1K. **b**, The same comparison is summarized per trajectory by its deepest writable block. **c**, Among the same 2K failures, deeper writeability predicts correction at 3K. **d**, On fixed persistent identities, downstream B13/B14 access can close and reopen across nearby checkpoints; curves show cohort medians.

B.4 Boundary-phase controller ablations

The two-dimensional view in Figure 3b is a fit-only phase-aligned projection inside the frozen top-four route. After fitting the joint raw, uncentered PCA on 128 fit differences, we fit within each target direction

$$z(\theta^*) = \mu + a \cos \theta^* + b \sin \theta^*$$

in four-dimensional route coordinates and orthonormalize the span of $[a, b]$. This frozen plane is then applied to 64 disjoint held-out pairs per direction. For the displayed run, held-out phase-plane R^2 is .935/.881; the corresponding unsupervised branch-local PC2–PC3 view gives .872/.875. Thus the panel shows that a phase-organized component exists within the causal top-four route, not that phase explains every top-four coordinate.

The executable controller in Figure 3 reads only the input-history boundary phase

$$\theta^* = \text{atan2}(-v^*/\omega_{\text{true}}, x^*)$$

and requested target direction $s \in \{+1, -1\}$. One pooled fit uses

$$[1, \cos \theta^*, \sin \theta^*, s, s \cos \theta^*, s \sin \theta^*] \rightarrow \hat{z}_{1:4}.$$

Because s has two values, this is algebraically equivalent to fitting a separate first-harmonic map $[1, \cos \theta^*, \sin \theta^*]$ for fast- and slow-target edits. Every model in Figure 11 is fit on the same 128 strict matched edits (64 per direction) and tested on 128 disjoint edits. Direction alone explains the common route offset, but not the state-varying coordinates; sharing one phase law across both targets also fails. The full target-specific phase model reaches held-out coordinate $R^2 = .828\text{--}.885$ across

three independently trained checkpoints. Adding explicit ω /amplitude features improves mean held-out R^2 by only .011, while raw boundary-state and larger derivative-shaped bases do not improve it.

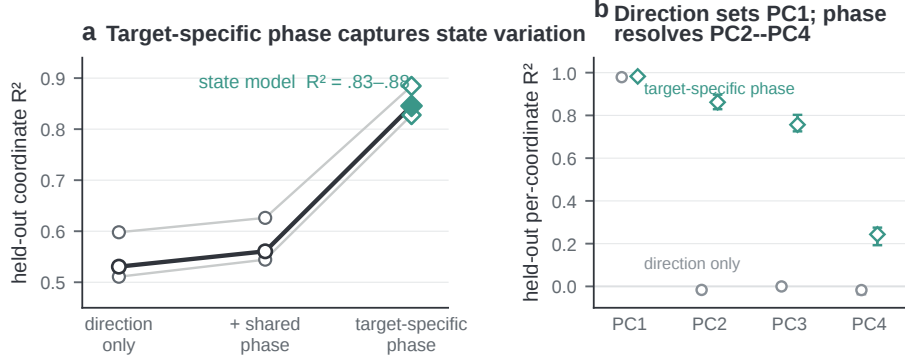


Figure 11: **Within the tested first-harmonic controller family, target-specific boundary phase is needed to predict the causal route coordinate.** **a**, Lines are independently trained checkpoints. Direction-only and a shared additive phase law leave substantial held-out error; allowing the first-harmonic coefficients to depend on target direction recovers the full state model. **b**, Direction alone predicts the coarse PC1 translation but almost none of PC2-PC4; target-specific phase predicts those state-varying coordinates. All values are fit-only held-out coordinate predictions, not decoded recovery rates.

Repeating the same fit-only QR-plane analysis on explicit 100K checkpoints replicates the structure at the localized functional layers B3, B6, and B4. Qualification is checkpoint-local; held-out fast/slow counts are 50/45, 60/53, and 54/53 for Runs A-C. Held-out fast/slow phase-plane R^2 is .868/.817 for Run A, .934/.881 for Run B, and .924/.895 for Run C (315 held-out matched-difference records total). For Run B, these values are nearly identical to the 50K result (.935/.881), whereas its branch-local PC2-PC3 control at 100K is .866/.824. The contrast shows that apparent loop tightness depends on the projection: the fit-only phase-aligned plane and unsupervised local PCs answer different questions within the same frozen route.

B.5 Boundary-state and after-conflict controllers

The boundary-state controller uses only the input boundary state and requested rewrite direction to predict $\hat{z}_{1.4}$ and reconstruct the condition-prefix edit through the frozen top-four basis. Baseline rollouts define the held-out strict-failure evaluation cohort but are not inputs to the controller. A complementary coordinate-difference model first decodes the natural conflict and estimates its realized frequency and future-start phase, $(\hat{\omega}_C, \hat{\phi}_C)$. Given a requested target state (ω_T, ϕ_T) , it forms

$$\Psi(\omega, \phi) = [\omega, \cos \phi, \sin \phi, \omega \cos \phi, \omega \sin \phi], \quad \Delta\Psi_{C \rightarrow T} = \Psi(\omega_T, \phi_T) - \Psi(\hat{\omega}_C, \hat{\phi}_C).$$

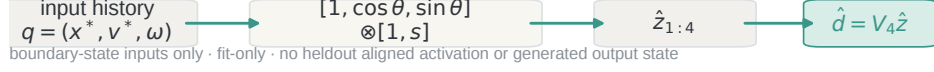
The requested physical frequency is mapped into the model’s decoded-frequency convention by a fit-only calibration, and target phase is supplied by a target-specific first-harmonic law fit on the same training split. On the 128 held-out receivers per run, constructing the edit reads only the natural conflict rollout—not the held-out aligned video or activation. Held-out coordinate R^2 is .828/.879/.850 across Runs A-C, and median decoded recovery over the six run-direction groups ranges from .895 to 1.004.

This difference form has useful structure. It is zero when target and current states agree, reverses sign when source and target are exchanged, and composes additively through intermediate states. The sine-cosine embedding removes the 2π phase discontinuity; $\Delta\omega$ supplies the coarse frequency translation, while the phase and ω -weighted phase terms provide position-like and velocity-like corrections. A single pooled linear map therefore supports continuous target specification, although the decoded evaluations in this paper are restricted to the two matched target families; they do not establish reliable control of unsupported middle frequencies. The model remains retrospective be-

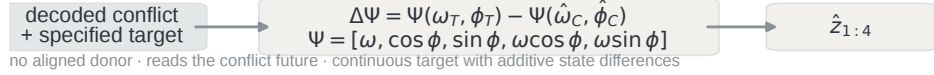
cause $(\hat{\omega}_C, \hat{\phi}_C)$ must be measured from the first generated future; the target state is specified without another donor video.

a Two physical parameterizations obey different information contracts

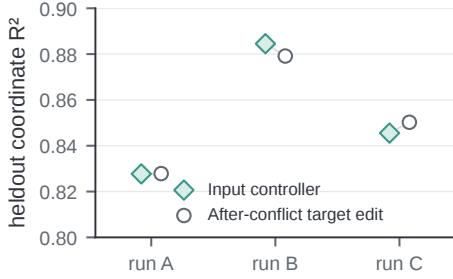
Executable input-state controller



Post-generation specified-target edit



b Comparable heldout coordinate fit



c Both parameterizations recover matched targets



Figure 12: **Two physical parameterizations describe the route under different information contracts.** **a**, The boundary-state controller uses only input boundary phase and target direction to construct its edit. The coordinate-difference model measures the decoded conflict state and compares it with a target state supplied by fit-only frequency calibration and phase prediction; it reads neither the held-out aligned video nor an aligned activation donor. **b**, Both parameterizations predict similar held-out route coordinates. **c**, Injecting either prediction recovers the matched target family, but only the boundary-state controller constructs its edit without reading the generated conflict future. Decoded interventions are evaluated only for the slow and fast matched target families, not unsupported middle frequencies.

B.6 Pendulum replication of physically organized editability

We repeat the behavior, edit-geometry, decoded-recovery, and layer-localization analyses in the Pendulum frequency_color_circle task. The frozen contract is summarized in Table 4.

Evidence-dependent behavior. All $4 \times 64 \times 11 = 2,816$ decoded rollouts have valid dynamics and appearance measurements, with physical-state identity—not individual hue rollout—as the clustering unit. Endpoint cues favor their training-associated frequency, whereas ambiguous cues expose stronger control by physical history; the generated future appearance changes with the selected motion rather than merely copying the input cue (Figure 13).

Physical-state coordinate geometry. We fit raw, uncentered PCA on fit matched differences only and retain the true top-four causal edit coordinates. For each target direction separately, a first-harmonic law of boundary phase—equivalently, boundary angle and angular velocity—defines a fit-only two-dimensional plane. Projecting the disjoint held-out edits into that frozen plane gives $R^2 = .975$ for target-low and $R^2 = .985$ for target-high ($n = 32$ each; Figure 14). The displayed loops are therefore geometry of activation edits, rather than a circle constructed from normalized physical inputs.

Table 4: **Pendulum replication contract and provenance.**

Component	Frozen setting
Model	Large frequency_color_circle, width 1152, seed 3407, Short/Long 50K 0dd907676adebe24fd44b1e8feae808a
Checkpoint	step 50,000; SHA256 2c7864615ac965a742e422f0cd721392
Behavior	64 physical states $\times$ 11 cues in each of four history-band cells; 2,816 decoded futures
Mechanism split	128 strict receivers; 64 fit and 64 disjoint held-out, balanced 32/32 per target direction
Intervention	after block 12 (zero based), all 20 flow-matching calls, 1,088 condition-prefix tokens only; target suffix unchanged
Evaluator	One appearance-tolerant geometry/frequency contract applied identically to natural, full, top-four oracle, fit-only, and layer-scan outputs
Same-rank comparison	top-four oracle coordinates versus top-four coordinates predicted by the direction-specific first harmonic; global scale 1.0 selected on fit only

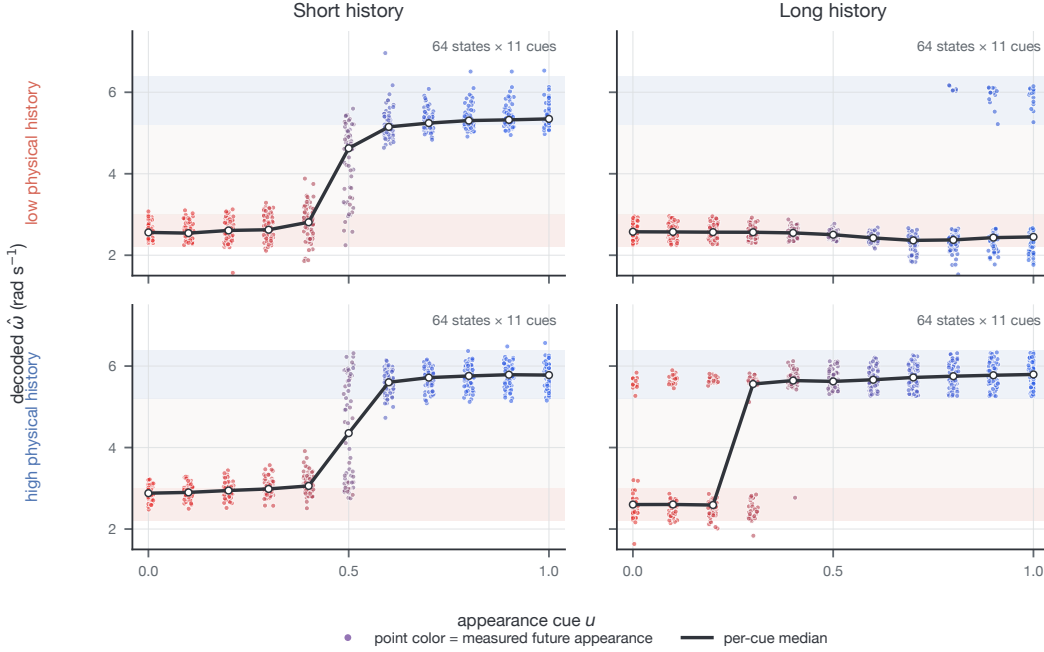


Figure 13: **Appearance cue and physical history compete in Pendulum decoded rollouts.** Rows hold low- or high-frequency physical history fixed, and columns compare Short with Long history. Each point is one decoded rollout at one of 11 input hues; point RGB is the measured appearance of the generated future. Black curves show the per-hue median decoded frequency, and pale horizontal bands mark the two training-supported frequency ranges.

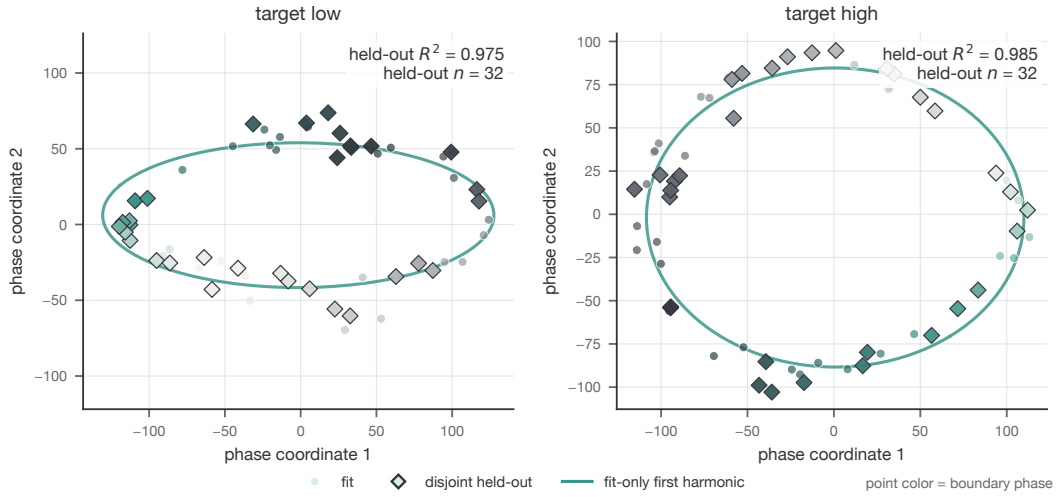


Figure 14: **Pendulum physical state organizes true causal edit coordinates.** Target-low and target-high edits are fit separately in the frozen top-four coordinate space. Circles are fit receivers, diamonds are disjoint held-out receivers, point color is boundary phase, and the thin curve is the fit-only first harmonic. Each facet reports held-out phase-plane R^2 .

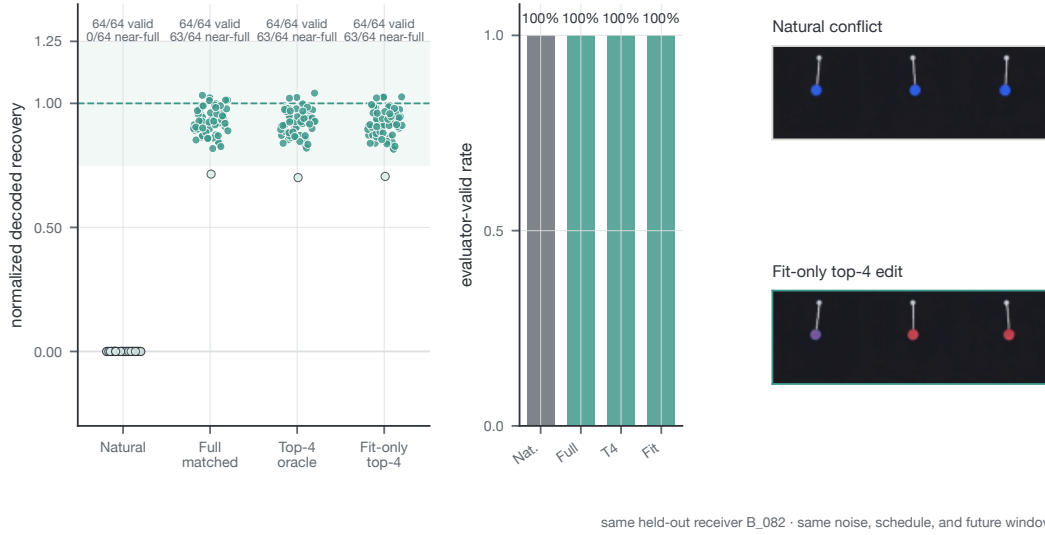


Figure 15: **A donor-free top-four Pendulum controller approaches the top-four oracle reference.** Every point is one of 64 receivers; the oracle and fit-only conditions use the same frozen top-four coordinate space. The side panel reports current evaluator validity for every condition, and labels report all-cohort near-full counts. The real decoded strips show the same held-out receiver before and after the donor-free edit.

Donor-free decoded recovery. All 64 held-out receivers are reported for natural conflict, full matched, top-four oracle, and fit-only top-four conditions, and all four conditions use the same evaluator. All 64/64 outputs are evaluator-valid. Their all-cohort near-full counts are 0, 63, 63, and 63, with median recovery 0, .925, .926, and .917, respectively. Thus the donor-free top-four controller approaches its same-rank top-four oracle reference: the oracle yields 57 physics-following futures and seven compromises, versus 58 and six for fit-only. No held-out donor activation or held-out edit norm enters the fit-only edit. Receiver B_082 supplies the frozen valid-all-four frame strip; the edit changes the joint appearance–dynamics continuation under the same noise, schedule, and future window (Figure 15).

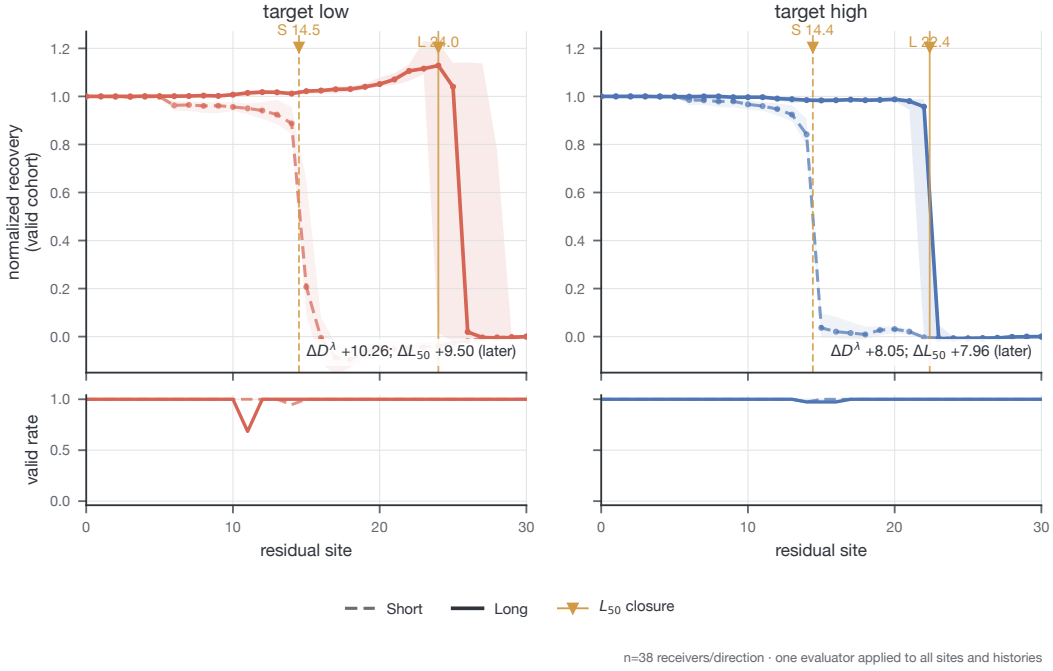


Figure 16: **History reshapes the Pendulum writeability profile.** Low- and high-target directions are shown separately with Short and Long common-strict curves overlaid. Thin amber lines mark separate L_{50} closure sites. Integrated writeability increases and L_{50} moves later in both directions. The lower axes report current evaluator validity only.

History reshapes the writeability profile. On the Short/Long common-strict cohort ($n = 38$ paired receivers per direction), Long increases the integrated depth D^λ in both directions: $+10.26$ sites for target-low and $+8.05$ for target-high. The midpoint L_{50} also moves later in both directions, by $+9.50$ sites for target-low and $+7.96$ sites for target-high. This replication licenses the canonical seed-3407 Pendulum result under the current evaluator, but not a Pendulum K/V mechanism claim, a non-periodic generalization, or a unified Pendulum–Projectile conclusion.

B.7 Persistent-error composition control

Figure 17 removes a possible sample-composition explanation for the training-time contraction in causal writeability. We freeze the trajectory identities that remain strict shortcut failures at all six evaluated checkpoints, then compare those same trajectories at 5K and 100K. Thus a later checkpoint cannot appear less writable merely because its local strict bank contains a different set of failures.

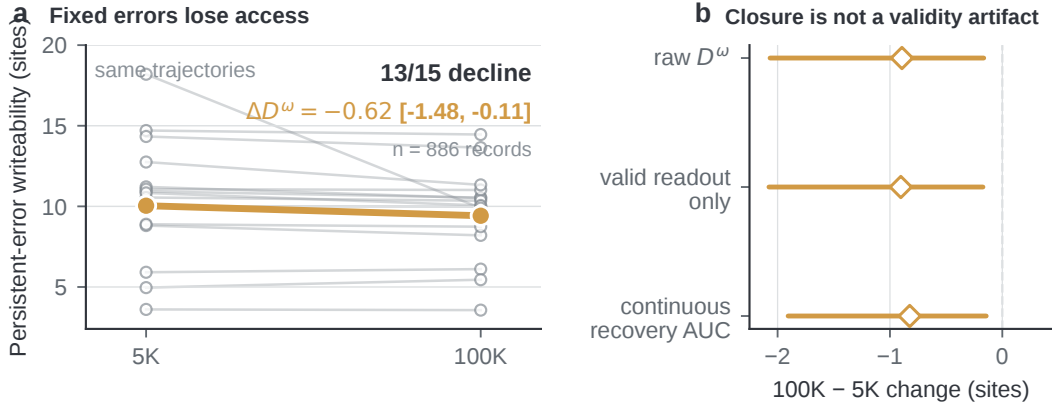


Figure 17: **The same unresolved shortcut errors become less causally writable during training.** **a**, Each gray line is one seed cohort evaluated on fixed persistent-error trajectory identities; amber is the trajectory-weighted mean. Thirteen of fifteen cohorts decline from 5K to 100K, with mean paired change $\Delta D^w = -0.62$ sites (seed-clustered 95% CI $[-1.48, -0.11]$). **b**, The same fixed cohort loses access under raw, valid-only, and continuous-recovery scoring.

B.8 Writeability robustness and history dependence

We test whether the writeability results depend on the near-full recovery threshold, measured output/state covariates, or invalid frequency readouts. We then report the operational Short/Long boundaries for the three matched-seed controls.

The checkpoint-centered cross-solution association is direction-asymmetric: matched fast-target writeability has $r = .44$ (95% CI $[.05, .71]$), whereas the matched slow-target estimate is $r = .29$ (95% CI $[-.03, .51]$). The pooled $r = .40$ association in Figure 4b summarizes an overall tendency, with a stronger estimate in the fast branch.

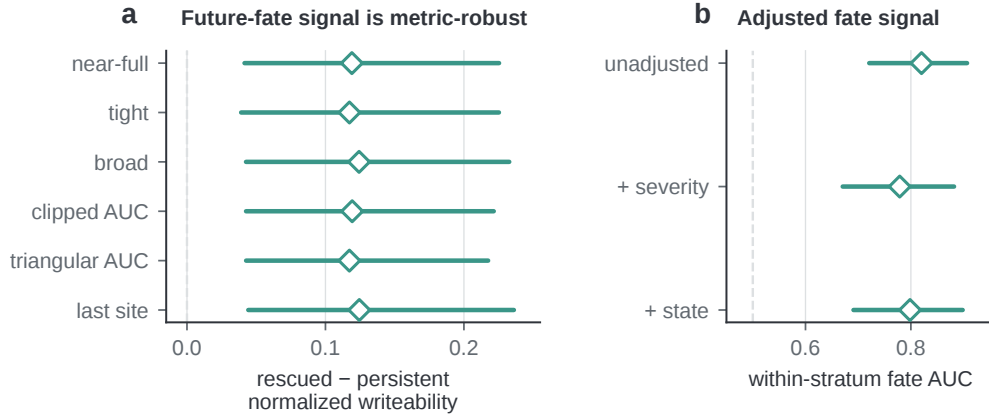


Figure 18: **Early writeability and training-time closure survive alternative definitions and measured controls.** **a**, The early gap between future-rescued and persistent failures remains under six definitions of writeability (157 rescued, 953 persistent). **b**, The same early separation remains after controlling natural error severity and boundary state. Diamonds are direction-balanced estimates; horizontal lines are seed-clustered trajectory bootstrap 95% intervals.

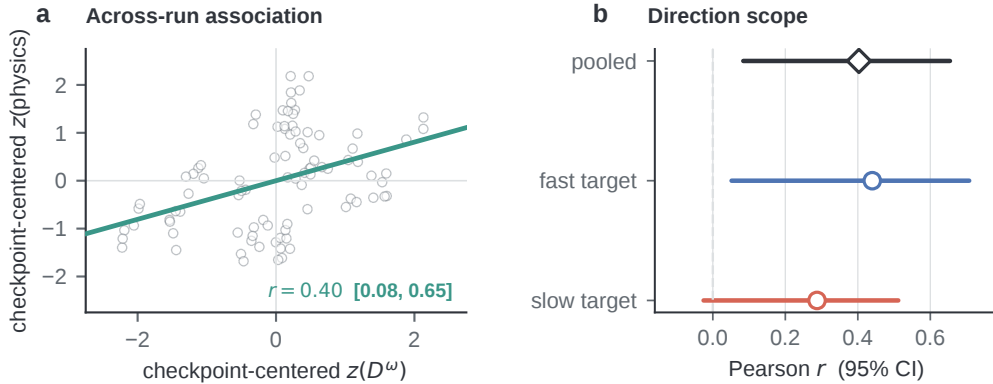


Figure 19: **Across-run writeability is associated with physics behavior at matched checkpoints.** **a**, After checkpoint centering, runs with larger D^ω tend to be more physics-following. **b**, The association is clearer for matched fast-target quantities than for the slow target; the pooled result summarizes a direction-asymmetric tendency. Complete six-checkpoint seed trajectories are the bootstrap unit.

Table 5: **Long checkpoints have later operational fast-target write boundaries in three matched-seed 50K comparisons.** Boundaries are the last after-block sites whose strong-rewrite rate is at least .75. Short and Long use the same architecture, step, and intervention protocol, but each checkpoint is evaluated on its own 128-pair strict bank; trajectory identities are not paired across history regimes.

Run	Short boundary	Long boundary	Long – Short
A	B3	B16	+13
B	B6	B8	+2
C	B6	B10	+4

B.9 Early error fate and downstream target write

Figure 20 asks whether trajectories with different future training outcomes already differ in downstream route realization at the same early checkpoint and boundary.

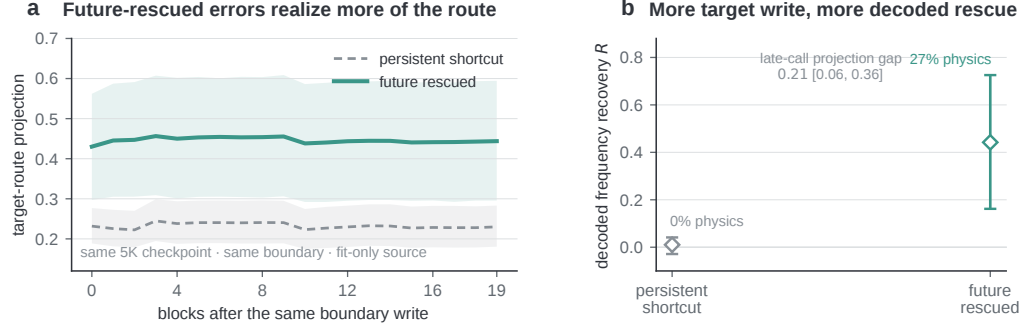


Figure 20: **Early failures later rescued by training already realize more of the condition-to-target route.** At the same 5K boundary, trajectories later rescued by training already show a larger route-aligned target-state response than persistent failures (a). A K-only intervention also changes decoded frequency only in the future-rescued group (b; 11 rescued and 12 persistent fast-target failures). The source controller is fit on separate trajectories.

B.10 Cross-run transfer controls and target-conditioned outcomes

The scale-plus-orthogonal maps in Figure 6b are fit on 128 matched differences and evaluated on 128 disjoint physical identities for every directed run pair. Matched within-direction coordinate R^2 ranges from .955 to .997, whereas shuffling trajectory identity within direction gives means of .066–.152; none of 20,000 shuffles reaches the matched value ($p < 0.00005$ for every map). This shuffle tests coordinate geometry, not decoded causal efficacy. The separate intervention test passes held-out boundary state through the source controller, frozen cross-run map, and target basis. All six directed maps redirect decoded frequency with pooled direction-median recovery $R = .942$ –.991.

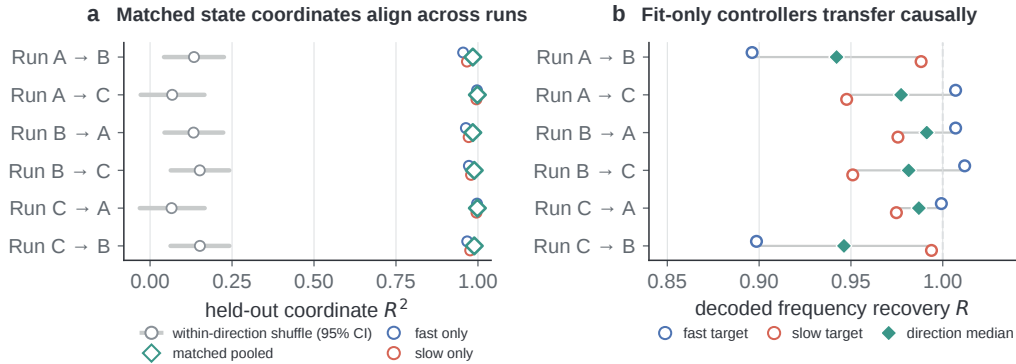


Figure 21: **Matched causal geometry transfers across independently trained models.** a, A fit-only scale and four-dimensional orthogonal map aligns held-out matched coordinates in all six directed run pairs; within-direction trajectory shuffles do not. b, Source-controller predictions transferred through the frozen maps recover decoded frequency in both rewrite directions. The shuffle in a is a coordinate null; the decoded claim in b comes from separate held-out interventions.

Absolute clean-P rates remain bounded by the target model’s own matched-edit ceiling; normalized recovery is therefore the primary cross-run transfer endpoint.

B.11 Oracle-coordinate compatibility across training checkpoints

We next ask whether a route coordinate selected at one training checkpoint can be expressed at another checkpoint in the same run. The assay uses exactly 5K, 10K, 20K, 50K, 80K, and 100K. Every checkpoint independently qualifies strict failures from the same frozen 256-trajectory library. Each source–target map is fit on the qualified intersection inside one permanent fit split and evaluated on the corresponding intersection inside the globally disjoint held-out split; a physical identity never changes split.

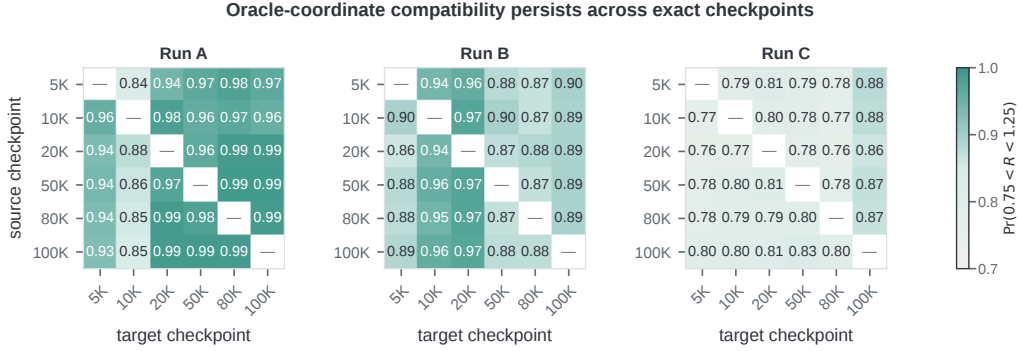


Figure 22: Oracle-selected route coordinates remain compatible across exact training checkpoints. Cells report the held-out fraction satisfying $.75 < R < 1.25$ after a source-oracle top-four coordinate is mapped by a fit-only scaled-orthogonal transform and injected at the target checkpoint’s own rewrite site. All 90 directed off-diagonal maps are shown; each cell contains 88–123 held-out physical trajectories, with Wilson 95% intervals in the source data. Blank diagonals mean self-transfer was not evaluated. The assay tests within-run temporal compatibility, not the non-oracle controller transfer in Figure 6b.

Compatibility is high but solution-dependent: the median cell fraction is .966, .893, and .796 for Runs A–C, respectively. This does not contradict contracting writeability. Figure 22 asks whether an oracle-selected coordinate can be transported between each checkpoint’s own rewrite site on a qualified intersection; D^ω asks how long a local condition-side edit remains directly effective across depth.

B.12 Shared mediation and implementation multiplicity

Having established shared route geometry, we compare separately calibrated head- and component-level interventions. The goal is to identify how causal responsibility is distributed within each run, not to rank components at a common gain.

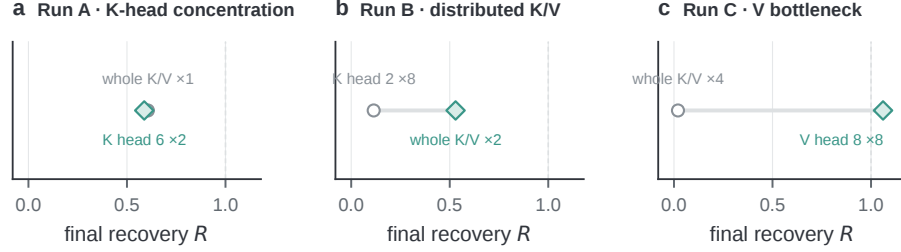


Figure 23: **Shared route semantics admit distinct downstream bottlenecks under the tested interventions.** In Run A, one K head closely reproduces the calibrated whole-K/V response. In Run B, the tested K-head intervention at gain 8 remains below whole-K/V at gain 2. In Run C, the tested whole-K/V gains 1, 2, and 4 do not cross the margin, whereas the selected V-head intervention does. Markers show mean final recovery over 16 held-out trajectories per panel. The panels establish implementation multiplicity. Independently calibrated doses do not support equal-dose component rankings.

Table 6: **Component restoration localizes a common attention-mediated target write.** The first four rows restore natural-conflict condition components inside the successful matched edit; effects are relative to the intact edit over six run-direction groups with 16 paired receivers each. The final rows compare post-attention target-state and MLP-output interventions. Intervals are hierarchical-bootstrap 95% CIs over group medians.

Paired contrast	Unit	ΔR	95% CI
Restore conflict Q – matched edit	6×16	+ .0004	[−.0010, +.0047]
Restore conflict K – matched edit	6×16	−.917	[−.967, −.028]
Restore conflict V – matched edit	6×16	−.003	[−.456, +.005]
Restore conflict K+V – matched edit	6×16	−.931	[−.985, −.908]
Edited post-attention target – edited MLP output	23 receivers / 6 groups	+ .960	[+.253, +1.024]
Restore conflict post-attention target – restore conflict MLP output	24 receivers / 6 groups	−.979	[−1.042, −.938]

B.13 Specificity and same-cohort full-edit references

For the Run-C V/h8 bottleneck, we test whether decoded rescue specifically tracks realization of the target route. We also compare the same 48 failures under the B9 head intervention and three B4 residual-edit references while keeping their distinct operators explicit.

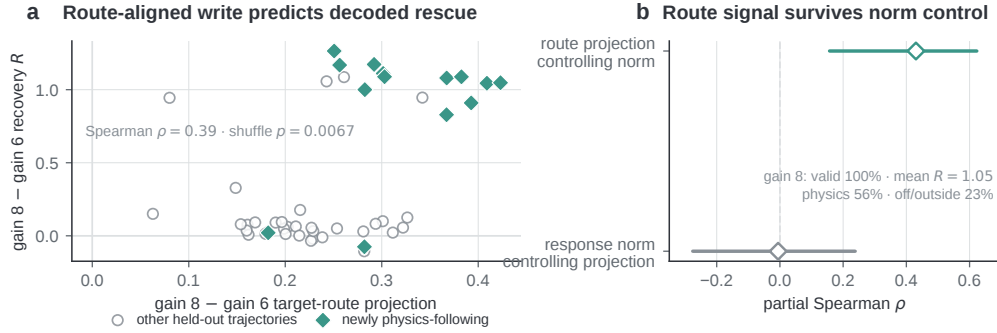


Figure 24: **Decoded rescue follows the route-aligned target write, not generic activation growth.** **a**, On the same 48 trajectories, the gain-6 to gain-8 increase in target-route projection tracks frequency recovery and identifies newly physics-following rollouts. **b**, Route projection remains associated with recovery after controlling response norm; the estimated partial association of response norm after controlling route projection is near zero ($\rho = -.006$, 95% CI $[-.276, .238]$).

Table 7: **Single-head rescue is comparable with full-edit references on the same 48 failures.** All interventions use the same held-out Run-C fast-target cohort, but the first three act on the B4 condition residual, whereas the single-head intervention acts on attention V at B9. We report Clean P descriptively because these interventions act at different sites and through different operators. P/S/C/O follow the decoded evaluator labels.

Intervention	Site / operator	Mean R	Median R	P	S	C	O
Full matched difference	B4 condition residual	.992	1.002	50.0%	0.0%	39.6%	10.4%
Top-four projection	B4 condition residual	.992	1.003	41.7%	0.0%	41.7%	16.7%
Fit-only state edit	B4 condition residual	.972	.994	43.8%	0.0%	41.7%	14.6%
Single V/h8, gain 8	B9 attention V head	1.048	1.087	56.3%	0.0%	20.8%	22.9%

B.14 FM-time allocation and decoded single-head rescue

Finally, we distribute the same gain-8 V/h8 write over different flow-matching-call windows and inspect both the immediate target-route response and the decoded continuation.

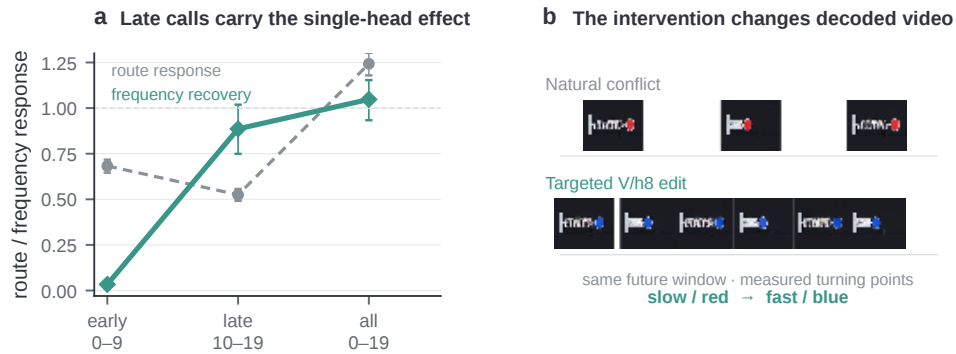


Figure 25: **The localized V-head write changes decoded video when applied over late flow-matching calls.** **a**, On the same 48 selection-clean Run-C fast-target failures, an early-call write is internally visible but behaviorally ineffective, while late or all-call allocation rescues frequency. A separate same-identity fast-target cohort ($n = 16$) shows that neither five-call half of the late window rescues alone. **b**, A representative exact replay changes the decoded joint continuation from red/slow to blue/fast; it is the robust- L_1 medoid selected from 28 eligible successful replays, and frames mark measured turning points in the same future window.